

(1001CJA100120004)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

12-07-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B,C,D	B,C	A,B,D	B,C	B,C,D	B,C,D	A,C,D	B	D	A
	Q.	11	12	13							
	A.	C	B	A							
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	5.00	8.00	3.00	5.00					

PART-2 : CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	A,B,D	A,B,C	A,B,D	A,C	A,B	A,D	A,B,C,D	C	A	D
	Q.	11	12	13							
	A.	B	C	D							
SECTION-II	Q.	1	2	3	4	5					
	A.	5.00	4.00	2.50	1.66 or 1.67	4.83					

PART-3 : MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	A,B,D	A,B,C	A,B,C	A,C	B,D	A,D	B,D	C	B	D
	Q.	11	12	13							
	A.	A	B	C							
SECTION-II	Q.	1	2	3	4	5					
	A.	1.00	1.00	4.00	4.00	4.00					

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	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	C	B	A	B	D	A	A,B,C,D	A,B,C,D	C
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B	B,C,D	A,B,C,D	A,D	A	B	B	C		

PART-2 : CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	D	C	B	D	C	D	A,C	B,D	A,B,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,D	B,D	B,D	A,B,C,D	D	A	B	B		

PART-3 : MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	B	B	C	C	C	D	A,C	A,D	B,C
	Q.	11	12	13	14	15	16	17	18		
	A.	B,C	B,C	A,B	A,B,C,D	A	D	C	A		



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PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (B,C,D)

Sol. Slope = $\frac{y}{x}$

$y = \text{slope} \times x$

= area under curve.

Slope is zero when $x = 10$

Slope is zero when it get maximum height

$$\text{So } H = \frac{1}{2} \times 10 = 5\text{m}, R = 10 \times 2 = 20\text{m}$$

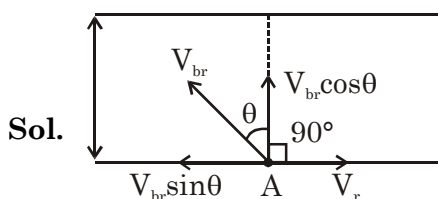
$$H = \frac{v^2 \sin^2 \theta}{2g} = 5$$

$$\frac{v^2 \sin^2 \theta}{g^2} = 1$$

$$\frac{v \sin \theta}{2} = 1 = \frac{T}{2}$$

So $T = 2$ sec.

2. Ans. (B,C)



Minimum drift cannot be zero as $V_r > V_{br}$
Condition for minimum drift is

$$\sin \theta = \frac{V_{br}}{V_r} = \frac{4}{5}$$

$$\Rightarrow \theta = 53^\circ$$

$$T_{\min} = \frac{d}{V_{br} \cos \theta}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{16}{25}} = 3/5$$

$$= \frac{36}{4 \times 3/5} = 15 \text{ sec}$$

$$(\text{drift})_{\min} = (V_r - V_{br} \sin \theta) \times T_{\min}$$

$$= (5 - 4 \times 4/5) \times 15$$

$$= \left(\frac{25 - 16}{5} \right) \times 15$$

$$= 27 \text{ m}$$

$$\text{Angle} = \alpha = 90^\circ + \theta = 90^\circ + 53^\circ = 143^\circ$$

3. Ans. (A,B,D)

Sol. $v^3 = 8x$

$$3v^2 \frac{dv}{dx} = 8$$

$$a = v \frac{dv}{dx} = \frac{8}{3v}$$

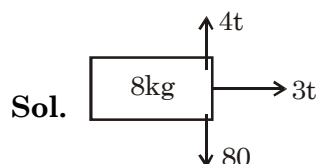
$$\text{if } v = 2, a = \frac{4}{3} \text{ m/s}^2$$

$$\text{if } v = 8, a = \frac{1}{3} \text{ m/s}^2$$

$$\text{if } x = 1 \Rightarrow v = 2 \Rightarrow a = \frac{4}{3} \text{ m/s}^2$$

$$\text{if } x = 8 \Rightarrow v = 4 \Rightarrow a = \frac{2}{3} \text{ m/s}^2$$

4. Ans. (B,C)



$$F_x = 3t = 3 \times 40 = 120$$

$$a_x = \frac{120}{8} = 15 \text{ m/s}^2$$

$$\vec{a} = 15\hat{i} + 10\hat{j}$$

$$a_x = \frac{3t}{8}$$

$$v_x = \int_0^{40} \frac{3t}{8} dt$$

$$= \frac{3}{16} \times 1600 = 300 \text{ m/s}$$

$$F_y = 4t - 80 = 80$$

$$a_y = \frac{80}{8} = 10 \text{ m/s}^2$$

$$a_y = \frac{4t - 80}{8}$$

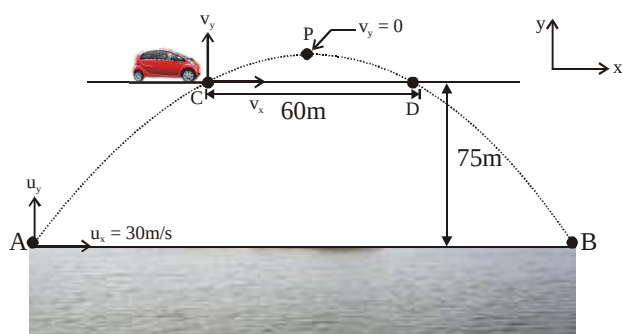
$$v_y = \int_{20}^{40} \left(\frac{4t - 80}{8} \right) dt$$

$$= \frac{1}{8} [2t^2 - 80t]_{20}^{40} = \frac{800}{8} = 100 \text{ m/s}$$

5. Ans. (B,C,D)

Sol. Time taken by car from C to D in x-direction is same as by stone from C to D.

$$\Rightarrow V_{\text{car}} = V_x \text{ of stone} = u_x = 30 \text{ m/s}$$



$$t_{C \text{ to } D} = \frac{60\text{m}}{30\text{m/s}} = 2\text{sec}$$

$$t_{C \text{ to } P} = t_{P \text{ to } D} = 1 \text{ sec}$$

at P; $v_y = 0$

we find v_y at C

$$\text{from C to P} \Rightarrow (v_y)_P = (u_y)_C + a_y t$$

$$\Rightarrow 0 = (u_y)_C - 10 \times 1 \Rightarrow (u_y)_C = 10 \text{ m/s}$$

For u_y at A, apply equation of motion

$$\text{from A to C} \Rightarrow (v_y^2)_C = (u_y^2)_A + 2a_y s_y$$

$$\Rightarrow (10)^2 = (u_y^2)_A - 2 \times 10 \times 75$$

$$\Rightarrow (u_y^2)_A = 100 + 1500 = 1600$$

$$\Rightarrow (u_y)_A = 40 \text{ m/s}$$

$$\Rightarrow u_x = 30 \text{ m/s} = u_y = 40 \text{ m/s}$$

$$\Rightarrow T = \frac{2u_y}{g} = \frac{2 \times 20}{10} = 8\text{sec}$$

$$\Rightarrow R = \frac{2u_x u_y}{g} = \frac{2 \times 40 \times 30}{10} = 240\text{m}$$

$$\text{at 1 sec, } v_y = 40 - 10 \times 1 = 30 \text{ m/s \& } v_x = 30 \text{ m/s}$$

$$\tan \theta = \frac{v_y}{v_x} = \frac{30}{30} = 1 \Rightarrow \theta = 45^\circ$$

$$\Rightarrow \text{Angle of projection} = \alpha$$

$$\Rightarrow \tan \alpha = \frac{u_y}{u_x} = \frac{40}{30} \Rightarrow \alpha = 53^\circ$$

6. Ans. (B,C,D)

$$\text{Sol. } \vec{v} = -\sin t \hat{i} + \cos t \hat{j} + 6t \hat{k}$$

$$|\vec{v}| = \sqrt{1 + 36t^2}$$

$$\vec{a} = -\cos t \hat{i} - \sin t \hat{j} + 6 \hat{k}$$

$$\vec{a} \cdot \vec{v} \text{ at } t = 0$$

$$(-\hat{i} + 6\hat{k}) \cdot (\hat{j}) = 0$$

$$\text{at } t = 0$$

$$\vec{v} = \hat{j}, \vec{a} = -\hat{i} + 6\hat{k} \text{ and } \vec{v} \perp \vec{a}$$

7. Ans. (A,C,D)

$$\text{Sol. (a) } \vec{F}_1 = \vec{F}_2 = 0$$

$$a_1 = a_2 = g \sin(\theta)$$

$$N = 0$$

$$(b) \vec{F}_1 = \vec{F}_2$$

$$\frac{F_1}{m_1} < \frac{F_2}{m_2} \quad \{m_1 > m_2\}$$

$$a_1 < a_2$$

$$\Rightarrow N \neq 0$$

$$\text{If } \frac{F_1}{m_1} = \frac{F_2}{m_2}$$

$$\Rightarrow a_1 = a_2$$

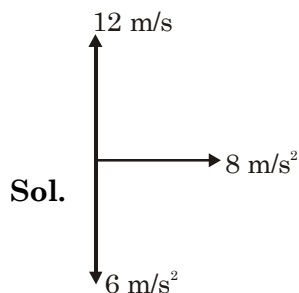
$$N = 0$$

8. **Ans. (B)**

9. **Ans. (D)**

10. **Ans. (A)**

11. **Ans. (C)**



$$u_y = 12, u_x = 0$$

$$a_y = -6, a_x = 8$$

$$t = 1, y = 12 \times 1 - \frac{1}{2} \times 6 \times 1^2 = 9$$

$$x = \frac{1}{2} \times 8 \times 1^2 = 4$$

$$v_x = 8 \times 1 = 8$$

$$v_y = 12 - 6 \times 1 = 6$$

$$v = \sqrt{6^2 + 8^2} = 10$$

12. **Ans. (B)**

Sol. $v_y = 12 - 6t = 0$

$$t = 2 \text{ sec}$$

$$x = 16 \text{ m}$$

$$y = 12 \text{ m/s}$$

$$v_x = 8 \times 2 = 16 \text{ m/s}, v_y = 0, v = 16$$

13. **Ans. (A)**

Sol. $-12 = 12 - 6t \Rightarrow t = 4 \text{ sec}$

$$x = 64 \text{ m}$$

$$y = 0$$

$$v_x = 32 \text{ m/s}$$

$$v_y = -12 \text{ m/s}$$

SECTION-II

1. **Ans. 2.00**

Sol. $V = 2/x$

$$\frac{dx}{dt} = \frac{2}{x}$$

$$x dx = 2 dt$$

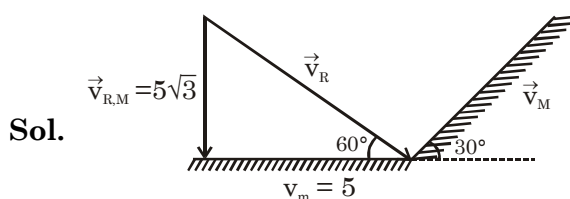
$$\frac{x^2}{2} = 2t$$

$$x = 2\sqrt{t} + C; C = 0$$

$$\frac{dx}{dt} = 2 \times \frac{1}{2\sqrt{t}}$$

$$v = \frac{1}{\sqrt{t}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

2. **Ans. 5.00**



$$\vec{v}_{R,M} = \sqrt{v_m^2 + v_{R,m}^2}$$

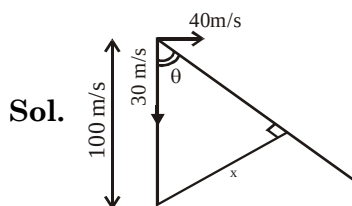
$$= 10 \text{ m/s}$$

On inlined plane $\vec{v}_R \perp \vec{v}_M$

$$\vec{v}_{R,M} = \vec{v}_R - \vec{v}_M$$

$$|\vec{v}_{R,M}| = \sqrt{10^2 + 5^2} = 5\sqrt{5}$$

3. **Ans. 8.00**



On taking relative motion first particle w.r.t. second.

$$a_{12} = g - g = 0$$

$$v_{12x} = 80 - 40 = 40 \text{ m/s}$$

$$v_{12y} = 0 - 30 = -30 \text{ m/s}$$

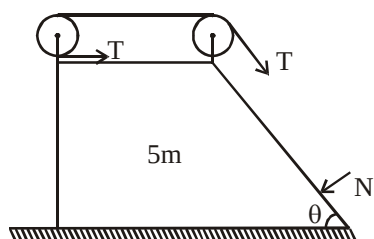
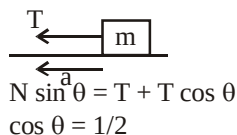
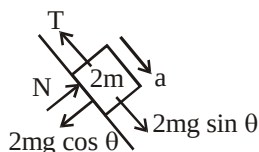
$$\tan \theta = \frac{40}{30} \Rightarrow \theta = 53^\circ$$

$$\sin \theta = \frac{x}{100} \Rightarrow x = \sin \theta \times 100 = \frac{4}{5} \times 100 = 80 \text{ meter.}$$

4. Ans. 3.00

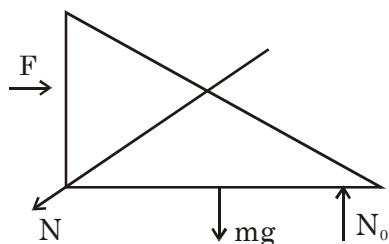
Sol. $2mg \sin \theta - T = 2ma$ (1)

$T = ma$ (2)



5. Ans. 5.00

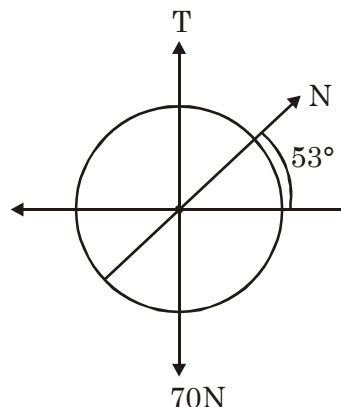
Sol From free body diagram of triangular block.



$$F = N \cos 53^\circ = N \sin 37^\circ \dots (i)$$

From the body diagram of sphere,

$$10 + N \sin 53^\circ = 70 \dots (ii)$$



from eq. (i) & (ii)

$$10 + \frac{F}{\sin 37^\circ} \sin 53^\circ = 70$$

$$\Rightarrow F = 60 \times \sin 37^\circ \times \frac{1}{\sin 53^\circ}$$

$$\Rightarrow F = 60 \times \frac{3}{5} \times \frac{5}{4} = 45N$$

$$\Rightarrow 9n = 45N$$

$$\Rightarrow n = 5$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A,B,D)

Sol. Atomic number of X = 25 [Mn]

	'X'		'Y'
	Mn		Mn ²⁺
Electronic configuration	[Ar]3d ⁵ 4s ²		[Ar]3d ⁵
μ =	$\sqrt{35}$ B.M.		$\sqrt{35}$ B.M.
Z_{eff} =	X	<	Y
d-electrons	5		5
Radius	X	>	Y

2. Ans.(A,B,C)

Sol. Correct order of radius.

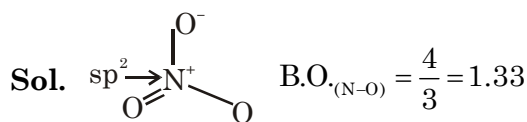
$$(A) S^{2-} > Cl^- > Ca^{2+} \left[\text{Radius} \propto \frac{1}{Z} \right]$$

$$(B) Li > B > N \text{ [Left to right in a period]}$$

$$(C) Cl^- > Na^+ > Mg^{2+} \left[\text{Radius} \propto \frac{1}{Z} \right]$$

(D) Ca > Mg > Cl [top to bottom (Ca > Mg) and left to right (Mg > Cl)]

3. Ans. (A,B,D)



$\Rightarrow \angle ONO = 120^\circ$ (because all hybrid orbitals are equivalent)

$\Rightarrow$ Nodal plane of π -bond lies in molecular plane

$\Rightarrow$ Nitrogen does not form five bonds

4. Ans.(A,C)

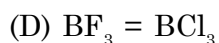
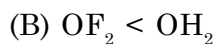
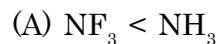
Sol. $Z = 17$ (Cl)

$\Rightarrow$ electron affinity of chlorine is highest

$\Rightarrow$ oxides of chlorine are acidic in nature

5. Ans.(A,B)

Sol. **CORRECT** order of bond angle.



6. Ans.(A,D)

Sol. (A) $p_x + p_x = \pi$ -bond

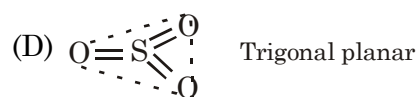
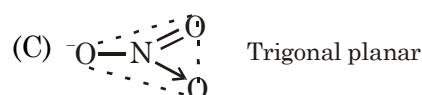
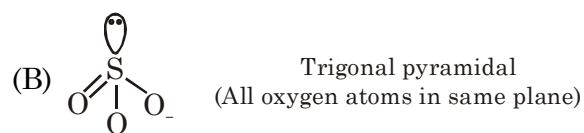
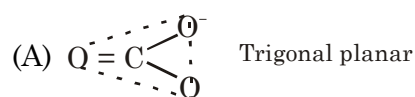
(B) $p_x + p_{xy} = \text{No bond}$

(C) $p_y + d_{xy} = \text{No bond}$

(D) $d_{xz} + d_{xz} = \pi$ bond

(If internuclear axis is z-axis)

7. Ans.(A,B,C,D)

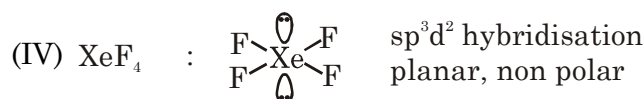
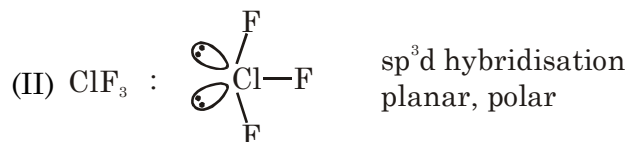
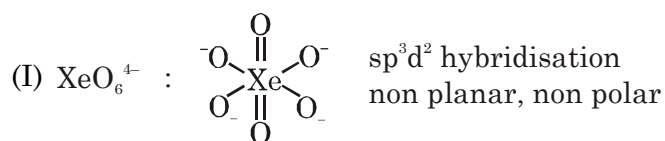


8. Ans.(C)

9. Ans.(A)

10. Ans.(D)

Sol.



11. Ans.(B)

12. Ans.(C)

13. Ans.(D)

SECTION-II

1. Ans.(5.00)

Sol. Se, Te, F, Cl, S having high electron affinity than oxygen.

2. Ans.(4.00)

Sol. NF_3 , PF_3 , BrF_3 , SF_2

3. Ans.(2.50)

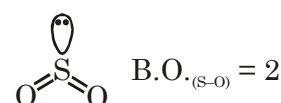
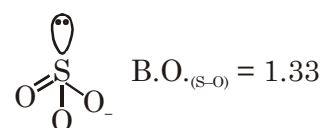
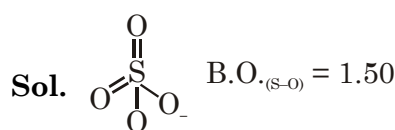
Sol. $\text{X} = \text{FeO}$, MnO , Na_2O , BaO are basic oxides
 $\text{Y} = \text{CO}$ is neutral oxide

4. Ans.(1.66 or 1.67)

Sol. $\text{X} = \text{XeF}_4$, ClO_4^- , XeO_3 , SF_6 , PCl_6^- having same bond length of all bonds.

$\text{Y} = \text{IF}_4^+$, ClF_3 , SF_4 does not have same bond length of all bonds

5. Ans.(4.83)



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,B,D)**

Sol. (A) $\sin A \cos B + \cos A \sin B = \sin (A + B)$
 (B) $\sin A \cos B - \cos A \sin B = \sin (A - B)$
 (C) $\cos A \cos B - \sin A \sin B = \cos (A + B)$
 (D) $\cos A \cos B + \sin A \sin B = \cos (A - B)$

2. **Ans. (A,B,C)**

Sol. 1 rad. $\approx 57^\circ$
 (A) $\sin 1 = \sin 57^\circ > \sin 1^\circ$
 (B) $\tan 2 = \tan 114^\circ$ (negative)
 (C) $\tan 1 = \tan 57^\circ > 0$
 (D) $\tan 2 = \tan 114^\circ < 0$

3. **Ans. (A,B,C)**

Sol. $f(1) = \cos 1 = +ve$
 ($\because \cos 1 \approx \cos 57.3^\circ > 0$)
 $f(2) = \cos 1 \cos 2 = -ve$
 (2^c lies in 2^{nd} quadrant)

$$f(3) = \underbrace{\cos 1}_{>0} \underbrace{\cos 2}_{<0} \underbrace{\cos 3}_{<0} = +ve$$

$$f(4) = \underbrace{\cos 1}_{>0} \underbrace{\cos 2}_{<0} \underbrace{\cos 3}_{<0} \underbrace{\cos 4}_{<0} = -ve$$

Similarly

$$f(5) \Rightarrow -ve ; f(6) = -ve ; f(7) = -ve$$

Hence $\lambda = 5$.

4. **Ans. (A,C)**

Sol. $T_r = \frac{1}{r^2} (1 + 2 + \dots + r)^2$
 $= \frac{1}{r^2} \frac{r^2(r+1)^2}{4} = \frac{r^2 + 2r + 1}{4}$

$$\Rightarrow T_7 = 16 \text{ and } S_{10} = \sum_{r=1}^{10} T_r = \frac{505}{4}$$

5. **Ans. (B,D)**

Sol. $(\log_\beta \alpha + \log_\alpha \beta)^2 - 2 = 142$
 $\Rightarrow \log_\beta \alpha + \log_\alpha \beta = \pm 12$

6. **Ans. (A,D)**

Sol. $a^{(\log_{11} 5)^2} = (a^{\log_{11} 5})^{\log_{11} 5} = (121)^{\log_{11} 5} = 25$

$$b^{(\log_3 49)^2} = (b^{\log_3 49})^{\log_3 49} = (\sqrt{3})^{\log_3 49} = 49^{\log_3 \sqrt{3}} = 7$$

$$c^{(\log_7 25)^2} = (c^{\log_7 25})^{\log_7 25} = (\sqrt{7})^{\log_7 25} = 5$$

7. **Ans. (B,D)**

Sol. Let $a = \frac{A}{r}$, $b = A$, $c = Ar$

$$b, c, a \text{ are in A.P.} \Rightarrow 2c = a + b$$

$$\Rightarrow 2Ar = \frac{A}{r} + A \Rightarrow 2r^2 - r - 1 = 0 \Rightarrow r = 1,$$

$$-\frac{1}{2}$$

$$\therefore \text{number are distinct so } r = -\frac{1}{2}$$

$$\Rightarrow a = -2A, b = A, c = -\frac{A}{2}$$

a, bc, abc are in H.P.

$$\frac{1}{a}, \frac{1}{bc}, \frac{1}{abc} \text{ are in A.P.}$$

$$\Rightarrow \frac{2}{bc} = \frac{1}{a} + \frac{1}{abc} \Rightarrow -\frac{4}{A^2} = \frac{1}{A^3} - \frac{1}{2A}$$

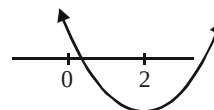
$$\Rightarrow -\frac{4}{A^2} = \frac{2-A^2}{2A^3} \Rightarrow -8A = 2-A^2 \Rightarrow A^2 - 8A - 2 = 0$$

$$A = \frac{8 \pm \sqrt{64+8}}{2} = \frac{8 \pm \sqrt{72}}{2} = (4 \pm 3\sqrt{2})$$

Matching list type (Q.8, Q.9, Q.10)

Sol. $f(x) = x^2 - (m-3)x + m$,

(I) exactly one root lies in $(0,2)$



$$f(0) \cdot f(2) < 0$$

$$\Rightarrow m(4 - 2m + 6 + m) < 0$$

$$\Rightarrow m(10 - m) < 0$$

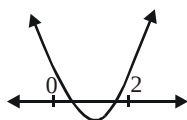
$$(-\infty, 0) \cup (10, \infty)$$

for $m = 0$, $x^2 + 3x = 0$, $x = 0, -3$, so $m \notin \{0\}$

for $m = 10$, $x^2 - 7x + 10 = 0$, $x = 5, 2$ so $m \notin \{10\}$

$$m \in (-\infty, 0) \cup (10, \infty)$$

(II) both roots lies in (0,2)



$$D \geq 0$$

$$(m-3)^2 - 4m \geq 0$$

$$\Rightarrow m^2 - 6m + 9 - 4m \geq 0$$

$$\Rightarrow m^2 - 10m + 9 \geq 0$$

$$\Rightarrow m \in (-\infty, 0] \cup [9, \infty) \quad \dots(1)$$

$$0 < \frac{-b}{2a} < 2 \Rightarrow 0 < \frac{(m-3)}{2} < 2$$

$$\Rightarrow 0 < m - 3 < 4 \Rightarrow m \in (3, 7) \quad \dots(2)$$

from (1) and (2) $m \in \phi$

(III) For both roots of $f(x)$ do not lie in (0,2)

All possible cases except when atleast one root lies in (0,2)

so $m \in (-\infty, 1] \cup [9, \infty)$ and $m \notin (-\infty, 0) \cup (10, \infty)$

so $m \in [0, 1] \cup [9, 10]$

(IV) Roots are opposite in sign

$$\Rightarrow f(0) < 0$$

$$\Rightarrow m < 0 \Rightarrow m \in (-\infty, 0)$$

For column-1

(I),(II) and (IV) are true and (II) is false

Column-2

$$g(x) = \log_2(x^2 - 8)$$

(i) $g(x) > 0 \Rightarrow \log_2(x^2 - 8) > 0$

$$\Rightarrow x^2 - 8 > 1 \Rightarrow x^2 > 9$$

$$\Rightarrow m \in (-\infty, -3) \cup (3, \infty)$$

(ii) $g(x) < 0 \Rightarrow \log_2(x^2 - 8) < 0$

$$\Rightarrow 0 < x^2 - 8 < 1$$

$$\Rightarrow x \in (-3, -2\sqrt{2}) \cup (2\sqrt{2}, 3)$$

(iii) $g(x) < 3 \Rightarrow \log_2(x^2 - 8) < 3$

$$\Rightarrow 0 < x^2 - 8 < 8$$

$$\Rightarrow x \in (-4, -2\sqrt{2}) \cup (2\sqrt{2}, 4)$$

(iv) $\log_2(x^2 - 8) \geq 3 \Rightarrow x^2 - 8 \geq 8$

$$\Rightarrow x^2 \geq 16,$$

$$\Rightarrow x \in (-\infty, -4] \cup [4, \infty)$$

for Column-2 :

(i), (ii) and (iv) are true (iii) is false.

For Column-3 :

$$t_r = \frac{1}{r(r+1)}$$

$$S_n = \sum_{r=1}^n t_r = \sum_{r=1}^n \frac{1}{r(r+1)}$$

$$= \sum_{r=1}^n \frac{1}{r} - \frac{1}{r+1}$$

$$= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$= 1 - \frac{1}{n+1}$$

for

(P) $S_{10} = 1 - \frac{1}{11} = \frac{10}{11}$

(Q) If $t_1 + t_2 + \dots + t_{10} < t_6 + t_7 + \dots + t_{15}$

$$\Rightarrow t_1 + t_2 + t_3 + t_4 + t_5 < t_{11} + t_{12} + t_{13} + t_{14} + t_{15}$$

$$\Rightarrow \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{5 \times 6} < \frac{1}{11 \times 12} + \frac{1}{12 \times 13} + \dots + \frac{1}{15 \times 16}$$

which is false.

(R) $S_\infty = \left(1 - \frac{1}{n+1}\right)$ when $n \rightarrow \infty = 1$

(S) $\sum_{r=2}^{\infty} t_r = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$

$$= \frac{1}{2}$$

In column-3, P, R and S are correct Q is wrong.

8. Ans. (C)

9. Ans. (B)

10. Ans. (D)

Matching list type (Q.11, Q.12, Q.13)

Sol. For column-1

(I) Let 'd' be common difference

$$\begin{aligned} \sum_{i=1}^{2020} \frac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}} &= \sum_{i=1}^{2020} \frac{\sqrt{a_{i+1}} - \sqrt{a_i}}{a_{i+1} - a_i} \\ &= \frac{1}{d} \sum_{i=1}^{2020} \sqrt{a_{i+1}} - \sqrt{a_i} \\ &= \frac{1}{d} (\sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{2021}} - \sqrt{a_{2020}}) \\ &= \frac{1}{d} \sqrt{a_{2021}} - \sqrt{a_1} = \frac{1}{d} \frac{a_{2021} - a_1}{\sqrt{a_1} + \sqrt{a_{2021}}} \\ &= \frac{2020d}{d\sqrt{a_1} + \sqrt{a_{2021}}} = \frac{2020}{\sqrt{a_1} + \sqrt{a_{2021}}} \end{aligned}$$

$$\begin{aligned} \text{III} \quad \sum_{r=1}^{2020} \frac{1}{a_r a_{r+1}} &= \frac{1}{d} \sum_{r=1}^{2020} \frac{a_{r+1} - a_r}{a_r a_{r+1}} \\ &= \frac{1}{d} \left(\sum_{r=1}^{2020} \left(\frac{1}{a_r} - \frac{1}{a_{r+1}} \right) \right) \\ &= \frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \dots + \frac{1}{a_{2020}} - \frac{1}{a_{2021}} \right) \\ &= \frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_{2021}} \right) = \frac{a_{2021} - a_1}{d(a_1)(a_{2021})} \\ &= \frac{2020d}{d a_1 a_{2021}} = \frac{2020}{a_1 a_{2021}} \end{aligned}$$

$$\text{IV} \quad \sum_{i=1}^{2019} \frac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}} = \frac{2019}{\sqrt{a_1} + \sqrt{a_{2020}}}$$

I, II and IV are correct

II is false.

For column-2

(i) $\log b_1, \log b_2, \log b_3, \dots$ are in A.P.

(ii) $b_1 = 2, b_2 = 1$

so common ratio $= \frac{1}{2}$

$$\sum_{i=1}^{\infty} b_i = \frac{b_1}{1-r} = \frac{2}{1-\frac{1}{2}} = 4$$

(iii) $b_1 = 1, b_3 = \frac{1}{4}$

$$\Rightarrow b_1 r^2 = \frac{1}{4} \Rightarrow r = \frac{1}{2} \quad (\text{Since } b_2 > 0)$$

(where r is common ratio)

$$\Rightarrow \sum_{i=1}^{\infty} b_i = \frac{b_1}{1-r} = 2$$

$$\text{(iv)} \quad \sum_{i=1}^n b_i = \frac{b_1(1-r^n)}{1-r}$$

$$\text{where } r = \frac{b_2}{b_1} = \frac{b_3}{b_2}$$

In column-2

(i), (ii) and (iv) are correct (iii) is false.

Column-3

$c_1^2, c_2^2, c_3^2, \dots, c_n^2$ are not always in G.P.

c_1, c_2, c_3 are in H.P.

$$\Rightarrow 2 \left(\frac{1}{c_2} \cdot \frac{1}{c_1} \right) = \frac{1}{c_3} + \frac{1}{c_1}$$

$$\Rightarrow \frac{c_1 - c_2}{c_1 c_2} = \frac{c_1 - c_3}{2c_1 c_3}$$

$$\Rightarrow \frac{c_1 - c_2}{c_1 - c_3} = \frac{c_2}{2c_3}$$

$$= \frac{c_1 c_2}{2c_1 c_3} = \frac{c_1 c_2}{c_2 c_3 + c_2 c_1}$$

$$\left(\frac{2}{c_2} = \frac{1}{c_1} + \frac{1}{c_3} \Rightarrow 2c_1 c_3 = c_2 c_3 + c_2 c_1 \right)$$

$$= \frac{c_1}{c_1 + c_3}$$

$$\Rightarrow \frac{c_1 - c_2}{c_1 - c_3} = \frac{c_1}{c_1 + c_3} \Rightarrow \frac{c_1}{c_1 - c_2} = \frac{c_1 + c_3}{c_1 - c_3}$$

(S) $c_1 = 1, c_2 = 3$

$$\Rightarrow 2c_1c_3 = c_2c_3 + c_2c_1$$

$$\Rightarrow 2c_3 = 3c_3 + 3 \Rightarrow c_3 = -3$$

In column-3

P, R and S are correct Q is wrong.

11. **Ans. (A)**

12. **Ans. (B)**

13. **Ans. (C)**

SECTION-II

1. **Ans. 1.00**

$$\frac{\cos x_1 + \cos x_2 + \dots + \cos x_{10}}{10} = |\operatorname{cosec} \alpha|$$

Maximum value of L.H.S. is 1 & minimum value of RHS is 1. possible when

$$\cos x_1 = \cos x_2 = \dots = \cos x_{10} = 1$$

$$x_1 = x_2 = \dots = x_{10} = 0$$

Harmonic mean

$$\begin{aligned} & \frac{10}{\left(\frac{1}{\cos x_1} + \frac{1}{\cos x_2} + \dots + \frac{1}{\cos x_{10}} \right)} \\ &= \frac{10}{(1+1+\dots+1)} = 1 \end{aligned}$$

2. **Ans. 1.00**

Sol. $P = \operatorname{cosec} \frac{\pi}{8} + \operatorname{cosec} \frac{2\pi}{8} + \operatorname{cosec} \frac{3\pi}{8}$

$$+ \operatorname{cosec} \left(2\pi - \frac{3\pi}{8} \right) + \operatorname{cosec} \left(2\pi - \frac{2\pi}{8} \right)$$

$$+ \operatorname{cosec} \left(2\pi - \frac{\pi}{8} \right) = 0$$

$$Q = \log \sin 10^\circ \cdot \log \sin 20^\circ \dots \log \sin 90^\circ \cdot \log \sin 100^\circ = 0$$

$$\text{Since } \log(\sin 90^\circ) = \log(1) = 0$$

$$\text{Hence } Q = 0$$

$$\therefore P + Q = 0.00$$

3. **Ans. 4.00**

Sol. $xz = y^2$ & $y^2 = \frac{x+z}{2}$

$$\Rightarrow \frac{x+z}{2} = xz$$

$$\text{A.M.} \geq \text{G.M.}$$

$$\Rightarrow xz \geq \sqrt{xz} \Rightarrow (xz)^2 \geq xz$$

$$\therefore xz \neq 0 \Rightarrow xz \geq 1$$

$$\Rightarrow y^2 \geq 1 \Rightarrow y \geq 1$$

$$\text{Now, } x + 2y + z = 2y^2 + 2y$$

$$\therefore (2y^2 + 2y)_{\min} = 2 + 2 = 4$$

4. **Ans. 4.00**

Sol. $\frac{\frac{x}{2} + \frac{x}{2} + y}{3} \geq \left(\frac{x^2y}{4} \right)^{\frac{1}{3}}$

$$\Rightarrow \frac{x^2y}{4} \leq 1 \Rightarrow x^2y \leq 4$$

5. **Ans. 4.00**

Sol. Let $a = \log_5 x$, $x > 1$ so $a > 0$

$$\log_x 0.04 = \log_x 5^{-2} = \frac{-2}{\log_5 x} = \frac{-2}{a}$$

$$\text{Now, } 2\log_5 x - \log_x 0.04 = 2a + \frac{2}{a}$$

$$\text{By A.M} \geq \text{GM (for } a > 0)$$

$$\frac{2a + \frac{2}{a}}{2} \geq \sqrt{2a \times \frac{2}{a}}$$

$$\Rightarrow 2a + \frac{2}{a} \geq 4$$

so least value is 4.

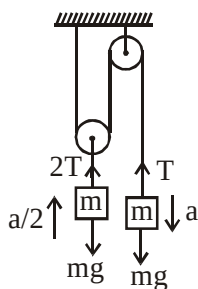
PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (B)****Sol.** $mg - T = ma$

$$2T - mg = m(a/2)$$

solving these, we get,

$$a = \frac{2g}{5}$$

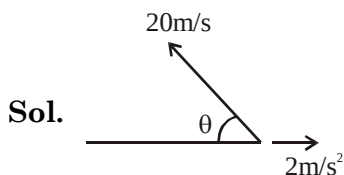
$$\text{so, } a_A = \frac{2g}{5}, a_B = \frac{g}{5}$$

**2. Ans. (C)****Sol.** $V = (\sin t) t^3$

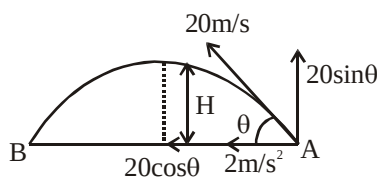
$$\frac{dv}{dt} = (3t^2) \sin t$$

$$t = \frac{\pi}{2}$$

$$\boxed{\frac{dv}{dt} = \frac{3\pi^2}{4} \text{ at } t = \frac{\pi}{2}}$$

3. Ans. (B)**Sol.**

w.r.t. train



$$H = \frac{(20 \sin \theta)^2}{2g}$$

$$T = \frac{20 \sin \theta}{g} \times 2$$

$$AB = 20 \cos \theta T + \frac{1}{2} \times 2T^2$$

4. Ans. (A)**Sol.** For coin w.r.t. lift

$$u = 0; a = -15 \text{ m/s}^2; s = -7.5 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

$$-7.5 = 0 + \frac{1}{2}(-15)t^2$$

$$t = \frac{15}{15} = 1$$

for coin w.r.t. ground

$$u = 10 \text{ m/s}; a = -g = -10 \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$s = 10 \times 1 + \frac{1}{2}(-10)(1)^2 = 5 \text{ m}$$

Positive so in upward direction.

5. Ans. (B)**Sol.** Time of flight will be same

$$u_A \sin \theta_A = u_B \sin \theta_B$$

$$\frac{R_A}{R_B} = \frac{u_A \cos \theta_A}{u_B \cos \theta_B}$$

$$\frac{R_A}{R_B} = \frac{\sin \theta_B \cos \theta_A}{\sin \theta_A \cos \theta_B} = \frac{\tan \theta_B}{\tan \theta_A}$$

6. Ans. (D)**Sol.** at angle θ

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$H_1 = \frac{u^2 \sin^2 \theta}{2g}$$

at $90 - \theta$

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$H_2 = \frac{u^2 \sin^2 (90 - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

$$4(H_1 H_2)^{1/2} = 4 \left(\frac{u^4 \sin^2 \theta \cos^2 \theta}{4g^2} \right)^{1/2}$$

$$= \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin 2\theta}{g} = R$$

7. **Ans. (A)**

Sol. Effective speed of the bullet = speed of bullet + speed of police jeep
 $= 180 \text{ m/s} + 45 \text{ km/h}$
 $= (180 + 12.5) \text{ m/s} = 192.5 \text{ m/s}$
 Speed of thief's jeep = $153 \text{ km/h} = 42.5 \text{ m/s}$
 Velocity of bullet w.r.t thief's car
 $= 192.5 - 42.5 = 150 \text{ m/s}$

8. **Ans. (A,B,C,D)**

Sol. $a_M = g$
 $a_m = 0$
 $\Rightarrow$ spring will extend.

9. **Ans. (A,B,C,D)**

Sol. $\frac{dx}{dt} = 2x + 1$

10. **Ans. (C)**

Sol. Mass (m) = 0.3 kg
 $\Rightarrow F = m \cdot a = 15x$

$$a = -\frac{15}{0.3}x = -\frac{150}{3}x = -50x$$

$$a = -50 \times 0.2 = 10 \text{ m/s}^2$$

11. **Ans. (A,B)**

Sol. $[A] = [L]$
 $[B] = [k] = [T^{-1}]$
 $[C] = [M^0 L^0 T^0]$

12. **Ans. (B, C, D)**

Sol. $\vec{A} + \vec{B} = -\vec{C}$
 $\vec{A} + \vec{B} + \vec{C} = 0$
 $\vec{C} + \vec{A} = -\vec{B}$
 $|\vec{C} + \vec{A}| = |\vec{B}|$

13. **Ans. (A,B,C,D)**

Sol. The displacement can be in any direction.

14. **Ans. (A, D)**

Sol. $a = \frac{F_1 - F_2}{(m_1 + m_2)}$

$$\Rightarrow N = F_2 + \frac{(F_1 - F_2)m_2}{m_1 + m_2}$$

$$\Rightarrow N = \frac{F_1 m_2 + F_2 m_1}{(m_1 + m_2)}$$

It must lie between F_1 & F_2

15. **Ans. (A)**

Sol. Velocity is maximum at $a = 0$

$$N = m_p g$$

16. **Ans. (B)**

Sol. $\frac{dv}{dt} = -\left(\frac{\alpha}{t_0/2}\right)t + \alpha$

$$v = -\left(\frac{2\alpha}{t_0} \times \frac{t^2}{2}\right) + \alpha t$$

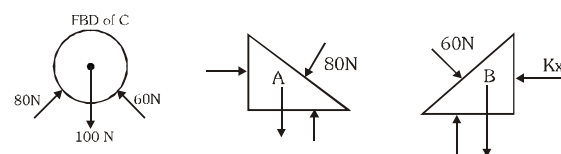
$$\frac{dx}{dt} = -\frac{\alpha}{t_0} t^2 + \alpha t$$

$$x = -\frac{\alpha t^3}{3t_0} + \frac{\alpha t^2}{2}$$

17. **Ans. (B)**

18. **Ans. (C)**

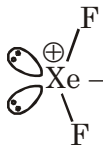
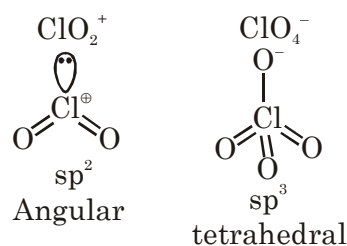
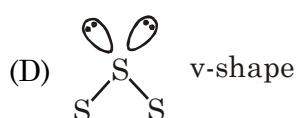
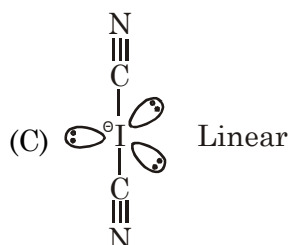
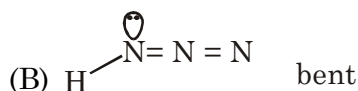
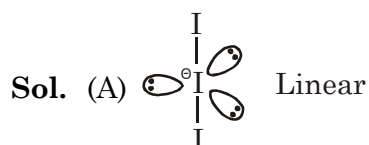
Sol. (17 & 18)



PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(B)****Sol.** (A) $(\text{Si}_2\text{O}_5)_n^{2n-} \rightarrow$ sheet silicate(B) $(\text{SiO}_3)_n^{2n-} \rightarrow$ Pyroxene (Chain silicate)(C) $\text{SiO}_4^{4-} \rightarrow$ Ortho silicate(D) $(\text{Si}_2\text{O}_7^{6-}) \rightarrow$ Pyro silicate2. **Ans.(D)****Sol.** (A) Carbon $\rightarrow \text{H}_2\text{CO}_3$ (dibasic)(B) Phosphorus $\rightarrow \text{H}_3\text{PO}_3$ (dibasic)(C) Sulphur $\rightarrow \text{H}_2\text{SO}_4$ (dibasic)(D) Chlorine $\rightarrow \text{HClO}, \text{HClO}_2, \text{HClO}_3, \text{HClO}_4$ (mono basic)3. **Ans.(C)****Sol.** Correct order of boiling point

(A) n-pentane > neo-pentane (due to branching)

(B) p-nitrophenol > o-nitrophenol (due to inter and intra molecular H bonding)

(C) $\text{NH}_3 < \text{HF} < \text{H}_2\text{O}$ (due to extent of H bonding)(D) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (due to molar mass)4. **Ans.(B)****Sol.** OF_4 and BF_6^- does not exist because central atom belongs to second period and unable to expand its octet. $\Rightarrow \text{PBr}_6^-$ does not exist due to crowding factor
 $\Rightarrow \text{XeF}_3^+ :$

 Bent 'T' shape
5. **Ans.(D)****Sol.** $X = 3 ; Y = 2$ 6. **Ans.(C)****Sol.** $\text{Cl}_2\text{O}_6(\text{s}) \rightarrow [\text{ClO}_2]^+ + [\text{ClO}_4]^-$ 7. **Ans.(D)****Sol.** In CCl_4 and SiCl_4 all repulsions are of same type. Both are sp^3 hybrid so bond angle is $109^\circ 28'$.8. **Ans.(A,C)**

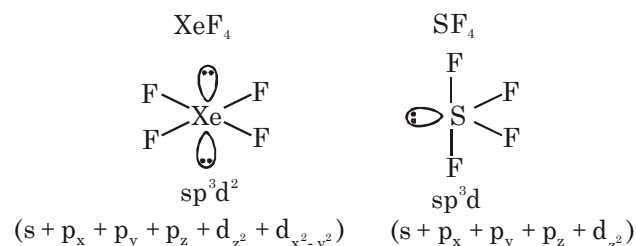
9. Ans.(B,D)

Sol. (A) Smallest anion is F^-

(C) Electronegativity : $Pb^{2+} < Pb^{4+}$

10. Ans.(A,B,D)

Sol.



$d_{x^2-y^2}$ is not commonly used.

11. Ans.(A,B,D)

Sol. $Z = 80$, element : Hg

Hg is not a transition metal

12. Ans.(B,D)

Sol. As per bent's rule

$$x_1 < x_2$$

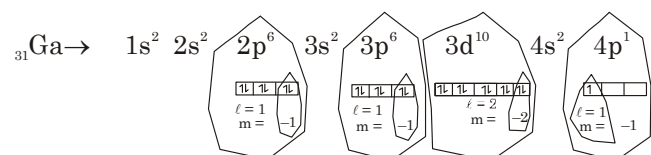
$\Rightarrow$ 'Cl' is bigger in size than 'F', so $y_1 < y_2$

13. Ans.(B,D)

Sol. Lattice energy $\propto \frac{|Q^+||Q^-|}{r_+ + r_-}$

14. Ans.(A,B,C,D)

Sol.



Maximum electrons having $l \times m = -1$ and -4 are 7.

(A) ${}_{13}Al \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^1$

Number of p electron = 7

(B) ${}_{19}K \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

Number of s electron = 7

(C) ${}_{27}Co \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^1$

Number of d electron = 7

(D) ${}_{89}Ac$ is present in 7th period

15. Ans.(D)

Sol. Correct order

(D) $F-F < Cl-Cl > Br-Br$

Due to inter electronic repulsion bond strength of F-F decreases.

16. Ans.(A)

Sol. (A) $(s+p_y)$ does not form any bond along x axis

(B) $p_y + p_y \rightarrow \pi$ -bond

(C) $d_{yz} + d_{yz} \rightarrow \delta$ -bond

(D) $d_{xy} + d_{xy} \rightarrow \pi$ -bond

17. Ans.(B)

Sol. I : Be, II : Ne, III : Mg, IV : N , V : F

$\Rightarrow$ Ne having highest ionisation energy

18. Ans.(B)

Sol. I : Be, II : Ne, III : Mg, IV : N , V : F

$\Rightarrow$ Order of size : $Be > N > F < Ne$
 $\quad \quad \quad \wedge$
 $\quad \quad \quad Mg$

PART-3 : MATHEMATICS

SOLUTION

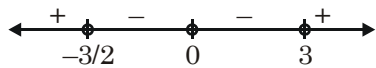
SECTION-I

1. **Ans. (B)**

Sol. $2x^4 - 3x^3 - 9x^2 > 0$

$$x^2(2x^2 - 3x - 9) > 0$$

$$x^2(2x + 3)(x - 3) > 0$$



$$x \in \left(-\infty, -\frac{3}{2}\right) \cup (3, \infty)$$

2. **Ans. (B)**

Sol. Let H = set of persons who know Hindi.

K = set of persons who know Kannada.

$n(H \cap K)$ = the number of persons who know both Hindi and Kannada is 70.

$$n(H \cup K) = n(H) + n(K) - n(H \cap K)$$

$$= 150 + 80 - 70 = 160$$

3. **Ans. (B)**

Sol. If $x > 1, y > 1, z > 1$ are in G.P.

$$\therefore y^2 = xz$$

$$\therefore 2 \ln y = \ln x + \ln z$$

$$\therefore \ln x, \ln y, \ln z \text{ are in A.P.}$$

$$\therefore 1 + \ln x, 1 + \ln y, 1 + \ln z \text{ are in A.P.}$$

$$\therefore \frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z} \text{ are in H.P.}$$

$$\Rightarrow \frac{1}{\ln ex}, \frac{1}{\ln ey}, \frac{1}{\ln ez} \text{ are in H.P.}$$

$$\text{or } \log_{ex} e, \log_{ey} e, \log_{ez} e \text{ are in H.P. } \mathbf{Ans.}$$

4. **Ans. (C)**

Sol. $5 - x^2 = \sqrt{5 - x}$

$$\text{assume } 5 = \lambda$$

$$(\lambda - x^2)^2 = \lambda - x$$

$$\lambda^2 - 2x^2\lambda + x^4 = \lambda - x$$

$$\lambda^2 - \lambda(2x^2 + 1) + x^4 + x = 0$$

$$\lambda = \frac{2x^2 + 1 \pm \sqrt{(2x^2 + 1)^2 - 4(x^4 + x)}}{2}$$

$$\lambda = \frac{2x^2 + 1 \pm \sqrt{4x^4 + 1 + 4x^2 - 4x^4 - 4x}}{2}$$

$$5 = \frac{2x^2 + 1 \pm \sqrt{(2x - 1)^2}}{2}$$

+sign

-sign

$$5 = \frac{2x^2 + 1 + 2x - 1}{2}$$

$$5 = \frac{2x^2 + 1 + 2x - 1}{2}$$

$$x^2 + x - 5 = 0$$

$$x^2 - x - 4 = 0$$

$$x = \frac{-1 \pm \sqrt{1 + 20}}{2}$$

$$x = \frac{1 \pm \sqrt{1 + 16}}{2}$$

$$x = \frac{-1 + \sqrt{21}}{2}, \frac{-1 - \sqrt{21}}{2} \text{ (rejected)}$$

$$x = \frac{1 - \sqrt{17}}{2}, \frac{1 + \sqrt{17}}{2} \text{ (rejected)}$$

number of solutions = 2

5. **Ans. (C)**

Sol. $4[(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)]$
 $- 6[(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta]$
 $= 4[1 - 3\sin^2\theta\cos^2\theta] - 6[1 - 2\sin^2\theta\cos^2\theta]$
 $= 4 - 12\sin^2\theta\cos^2\theta - 6 + 12\sin^2\theta\cos^2\theta = -2$

6. **Ans. (C)**

Sol. As $y = x^2 - 2ax + 1$

then the parabola's turning point is at the vertex $(a, 1 - a^2)$

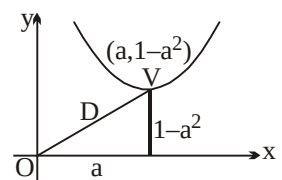
$$\left[-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right]$$

The distance D from the origin to this point is

$$D^2 = a^2 + (1 - a^2)^2$$

$$= a^4 - a^2 + 1$$

$$= \left(a^2 - \frac{1}{2}\right)^2 + \frac{3}{4}$$



so D^2 , and hence D , is at a minimum when

$$a^2 = \frac{1}{2}$$

7. Ans. (D)

Sol. $1, \frac{1}{p}, \frac{1}{q}, \frac{1}{4} \rightarrow \text{A.P.}$

$$\frac{1}{4} = 1 + (4-1)D \Rightarrow D = -\frac{1}{4}$$

$$\frac{1}{p} = 1 + D \Rightarrow p = \frac{4}{3}$$

$$\frac{1}{q} = 1 + 2D \Rightarrow q = 2$$

Hence, $(3p + q) = 6$ Ans.

8. Ans. (A,C)

Sol. $\frac{2x}{z} - \frac{3y}{z} = 1$ & $\frac{x}{z} + \frac{3y}{z} = 14$

Solving $\frac{x}{z} = 5$; $\frac{y}{z} = 3$

$$\Rightarrow \frac{\left(\frac{x}{z}\right)^2 + \frac{3x}{z} \cdot \frac{y}{z}}{\left(\frac{y}{z}\right)^2 + 1} = \frac{25 + 3(5)(3)}{(3)^2 + 1} = \frac{70}{10} = 7$$

9. Ans. (A,D)

Sol. Expression $= \log_{10} (\tan 6^\circ) + \log_{10} (\tan 12^\circ)$
 $+ \dots + \log_{10} (\tan 84^\circ)$

$$= \log_{10} ((\tan 6^\circ \cdot \tan 84^\circ) \cdot (\tan 12^\circ \cdot \tan 78^\circ) \cdot \dots \cdot (\tan 42^\circ \cdot \tan 48^\circ))$$

$$= \log_{10} ((\tan 6^\circ \cdot \cot 6^\circ) \cdot (\tan 12^\circ \cdot \cot 12^\circ) \cdot \dots \cdot (\tan 42^\circ \cdot \cot 42^\circ))$$

$$= \log_{10} 1 = 0$$

10. Ans. (B,C)

Sol. We have

$$x^2 + ax + b = 0 \begin{cases} \sec^2 \pi/8 \\ \operatorname{cosec}^2 \pi/8 \end{cases}$$

$$\text{Now, sum of roots} = \sec^2 \frac{\pi}{8} + \operatorname{cosec}^2 \frac{\pi}{8}$$

$$= \frac{1}{\cos^2 \frac{\pi}{8}} + \frac{1}{\sin^2 \frac{\pi}{8}} = \frac{\sin^2 \frac{\pi}{8} + \cos^2 \frac{\pi}{8}}{\sin^2 \frac{\pi}{8} \cdot \cos^2 \frac{\pi}{8}}$$

$$= \frac{1}{\sin^2 \frac{\pi}{8} \cdot \cos^2 \frac{\pi}{8}} = \sec^2 \frac{\pi}{8} \cdot \operatorname{cosec}^2 \frac{\pi}{8}$$

= Product of roots

$$\Rightarrow -a = b$$

$$\therefore a + b = 0$$

11. Ans. (B,C)

$$\text{Sol. L.H.S.} = \sum_{r=0}^{n-1} \log_2 \left(\frac{r+2}{r+1} \right)$$

$$= \log_2 \left(\frac{2}{1} \right) + \log_2 \left(\frac{3}{2} \right) + \log_2 \left(\frac{4}{3} \right) + \dots + \log_2 \left(\frac{n}{n-1} \right) + \log_2 \left(\frac{n+1}{n} \right)$$

$$= \log_2 \left(\frac{2}{1} \times \frac{3}{2} \times \frac{4}{3} \times \dots \times \frac{n}{n-1} \times \frac{n+1}{n} \right)$$

$$= \log_2 (n+1) \quad \dots (1)$$

$$\text{Now R.H.S.} = \prod_{r=10}^{99} \log_r (r+1) = \log_{10} 11 \times \log_{11} 12 \times$$

$$\log_{12} 13 \times \dots \times \log_{98} 99 \times \log_{99} 100$$

$$= \frac{\log 11}{\log 10} \times \frac{\log 12}{\log 11} \times \frac{\log 13}{\log 12} \times \dots \times \frac{\log 99}{\log 98} \times \frac{\log 100}{\log 99}$$

$$= \log_{10} 100 = 2 \quad \dots (2)$$

From (1) & (2), we get

$$\log_2 (n+1) = 2$$

$$\Rightarrow n+1 = 4$$

$$\therefore n = 3 \text{ Ans.}$$

12. Ans. (B,C)

Sol. (B) We have

$$3x^{\log_3 4} + 4^{\log_3 x} = 64$$

$$\Rightarrow 3 \cdot 4^{\log_3 x} + 4^{\log_3 x} = 64$$

$$\Rightarrow 4 \cdot 4^{\log_3 x} = 64$$

$$\therefore 4^{\log_3 x} = 16$$

$$\Rightarrow \log_3 x = 2$$

$$\therefore x = 9$$

$\Rightarrow$ square root of the root of the given equation = 3.

13. Ans. (A,B)

$$\text{Sol. As } 3 + 2\sqrt{2} = (\sqrt{2} + 1)^2$$

$$\text{and } 3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$$

$$\therefore \text{Expression} = \log_2 \left((\sqrt{2} + 1) + (\sqrt{2} - 1) \right)$$

$$= \log_2 (2\sqrt{2}) = \log_2 2^{3/2} = \frac{3}{2} = \frac{1}{6} = \frac{p}{q}$$

$$\therefore p + q = 7 \text{ Ans.}$$

14. Ans. (A,B,C,D)

Sol. $\ln(\tan x) = 0$

$$\tan x = e^0 = 1$$

$$\tan x = 1$$

now check options

Paragraph for Question 15 and 16

Sol. $f(x) = ax^2 + bx + c$

$f(5) = -3f(2)$ and $x = 3$ satisfies $f(x) = 0$

$$(i) \therefore 9a + 3b + c = 0 \quad \dots(1)$$

$$25a + 5b + c = -3(4a + 2b + c)$$

$$37a + 11b + 4c = 0 \quad \dots(2)$$

also $36a + 12b + 4c = 0 \quad \dots(3)$ [multiplying equation (1) by 4]

$$\underline{a - b = 0} \quad \Rightarrow a = b$$

$$\alpha + \beta = -\frac{b}{a} = -1, \quad \text{but } \alpha = 3$$

$$3 + \beta = -1 \quad \Rightarrow \beta = -4$$

hence the other roots is -4 **Ans.**

$$(ii) a + b + c = f(1)$$

$$\text{now } f(x) = a \left[x^2 - \left(-\frac{b}{a} \right) x + \frac{c}{a} \right] = a[(x - \alpha)(x - \beta)]$$

$$= a[(x - 3)(x + 4)] = a(x^2 + x - 12)$$

$$f(1) = -10a$$

$\Rightarrow$ can not be uniquely determined $\Rightarrow$ **(D)**

15. Ans. (A)

16. Ans. (D)

Paragraph for Question 17 and 18

17. Ans. (C)

$$Q_r = (r^2 + r + 1) - ((r - 1)^2 - (r - 1) - 1)$$

$$Q_r = (2r - 1) + 1 \Rightarrow 2r$$

$$V_r = 2r(2r + 2) - 2 \Rightarrow 4 \underbrace{r(r + 1)}_{\text{even number}} - 2$$

$V_r = 8N - 2 \Rightarrow$ Remainder when V_r divided by 8 is 6.

18. Ans. (A)

$$\prod_{r=1}^n \frac{S_r}{S_{r-1}} = \frac{S_1}{S_0}, \frac{S_2}{S_1}, \frac{S_3}{S_2}, \dots, \frac{S_n}{S_{n-1}} = \frac{S_n}{S_0}$$

$$S_0 = 1 \Rightarrow S_n$$